



Design Review Report

Kiln

kiln.zip

Report ID

3f567ddf-ddd1-4718-85ae-1f56b581ff32

Revision

Initial

Report Date

June 28, 2026

Generated

June 28, 2026

Project

Kiln

Netlist

kiln.zip

Model

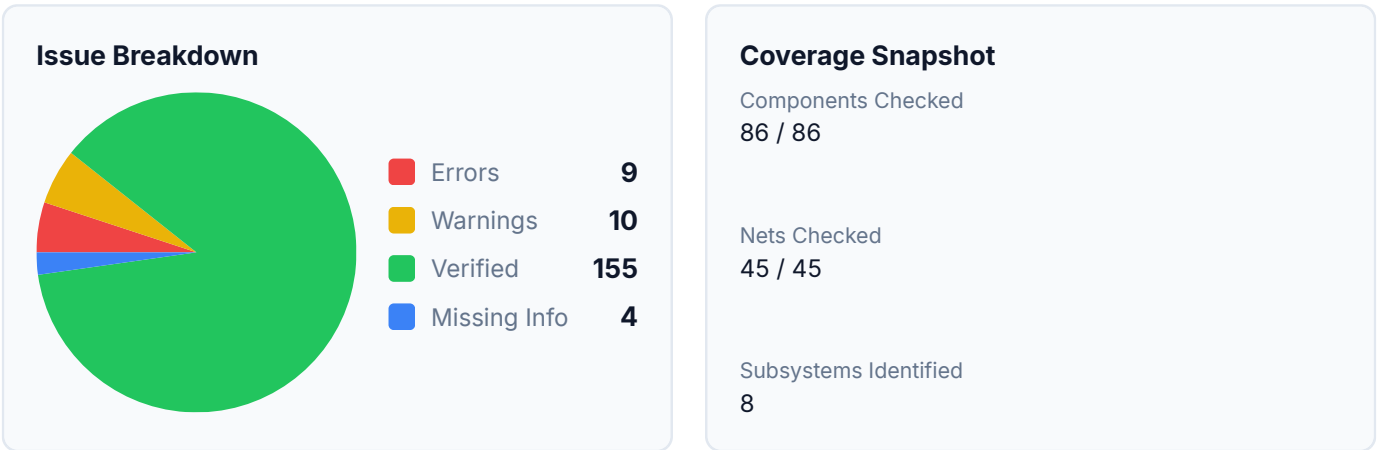
gemini-3-flash

Warning

AI can make mistakes. Findings are automated and should be validated by a qualified engineer before fabrication or release. Check important info.

Executive Summary

The system design utilizes a dual-processor architecture with an ESP32-S3 for main control and a Raspberry Pi Pico for safety functions, supported by isolated thermal and current monitoring. Critical hardware flaws were identified in the power and relay subsystems: the primary and safety relay circuits utilize latching relays with single-ended drivers that cannot perform a 'Reset' operation, and the contacts are wired to create direct 12V-to-ground short circuits when activated. The ESP32-S3 boot process is obstructed by pull-up resistors on strapping pins GPIO45 and GPIO46, which may force invalid flash voltages or boot modes. Additionally, the isolated data interface between processors contains signal direction mismatches and missing pull-up resistors on open-collector outputs, while the LDO regulators utilize ceramic capacitors that violate minimum ESR stability requirements.



Report Details

Report ID	3f567ddf-ddd1-4718-85ae-1f56b581ff32	Revision	Initial
Report Date	June 28, 2026 at 07:34 AM UTC	Generated	June 28, 2026 at 07:37 AM UTC
Model	gemini-3-flash	Provider	google

Datasheets Used

- LM1117T-3.3.pdf
<https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/2283/L...>
- EE2-12SNUH.pdf
https://yageogroup.com/content/datasheet/asset/file/KEM_R7002_EC2_EE2
- ESP32-S3-DevKitC-1 v1.1 - ESP32-S3 - — esp-dev-kits latest documentation.pdf
- MAX31856.pdf
<https://www.mouser.com/datasheet/3/1014/1/max31856evsys.pdf>
- TCMT1109.pdf
<https://www.vishay.com/docs/83510/tcmt11.pdf>
- RaspberryPi_Pico.pdf
<https://datasheets.raspberrypi.com/pico/pico-datasheet.pdf>
- SS16FP.pdf
<https://www.onsemi.com/pdf/datasheet/s110fp-d.pdf>
- BSP296NH6327XTSA1.pdf
https://www.infineon.com/dgdl/BSP296N_rev2+0.pdf

Errors (9)

1. [K1] Primary Relay Control Inoperability and Power Short

Full Description

Relays K1, K2, K3, and K5 (EE2-12SNUH) are single-coil latching variants that require polarity reversal to transition between 'Set' and 'Reset'. The current low-side MOSFET drivers (Q1, Q2) can only energize the coil in one direction. Furthermore, the relay contacts are wired such that they short the 12V_Main supply directly to the MOSFET drain (and subsequently to ground when active) in the default 'Reset' position. This creates a low-impedance short circuit that will likely damage the MOSFETs and relay contacts.

Recommendation

Replace the latching relays with non-latching (monostable) variants (e.g., EE2-12NUH) to work with single-ended drivers, or implement H-bridge drivers for polarity reversal. Isolate the relay contacts from the coil drive nets and connect the contact commons to the intended load source.

Affected Components

K1, K2, K3, K5, Q1, Q2

Affected Nets

12V_Main, GND_Main, Net - (D19-A2), Net - (D21-A2)

Datasheet Evidence

EE2-12SNUH.pdf (p.3)

"Single coil latch type (Reset position) ... 4-5 and 9-8 are connected. Coil polarity for Set: Pin 1 (+), Pin 12 (-)."

2. [K4] Safety Relay Reset Capability and Contact Short

Full Description

Similar to the primary relays, K4 is a single-coil latching relay driven by a single MOSFET (Q4), preventing a 'Reset' operation. Additionally, the NO contact (Pin 8) is connected to 12v_Safty and the Common (Pin 9) is connected to the control line (Net-(D26-A2)). Activating the relay will connect the 12V supply directly to the ground-referenced control line, causing a short circuit.

Recommendation

Replace K4 with a non-latching variant for fail-safe operation or use a bipolar driver. Rewire the contacts (Pins 8, 9, 10) to switch the external load independently of the control and power rails.

Affected Components

K4, Q4

Affected Nets

12v_Safty, Net - (D26-A2)

Datasheet Evidence

EE2-12SNUH.pdf (p.9)

"Only the specified polarity should be used to energize the coil."

3. [U18] ESP32-S3 Strapping Pin Conflict (GPIO45)

Full Description

With R35 (10k) pulling GPIO45 HIGH, the ESP32-S3 is forced to select a 1.8V internal supply for the SPI flash (VDD_SPI) at boot. Standard WROOM-1 modules typically require 3.3V; forcing 1.8V will prevent the flash from initializing and the system from booting.

Recommendation

Remove pull-up R35 from GPIO45. Relocate the 'thermoFault_1' signal to a non-strapping GPIO to avoid interference with the boot sequence.

Affected Components

U18, R35

Affected Nets

/MainController/thermoFault_1

Datasheet Evidence

ESP32-S3-DevKitC-1 v1.1 - ESP32-S3 - — esp-dev-kits latest documentation.pdf (p.7)

"GPIO45: VDD_SPI voltage selection"

4. [U18] ESP32-S3 Strapping Pin Conflict (GPIO46)

Full Description

GPIO46 must be sampled LOW during reset for a normal SPI flash boot. The pull-up R76 (10k) forces this pin HIGH, which may place the device into a reserved or invalid boot mode.

Recommendation

Remove R76 or change it to a pull-down resistor to ensure a logic LOW state during the reset sampling window.

Affected Components

U18, R76

Affected Nets

/MainController/thermoDrdy_1

Datasheet Evidence

ESP32-S3-DevKitC-1 v1.1 - ESP32-S3 - — esp-dev-kits latest documentation.pdf (p.6)

"GPIO46: Strapping pin (internal pull-down)"

5. [U13] Missing Pull-up Resistors on Open-Drain Status Pins

Full Description

Status pins ~FAULT and ~DRDY on U13 (MAX31856) are open-drain outputs. Without external pull-up resistors to the 3.3v_Safty rail, these signals will remain floating and produce undefined logic states at the Raspberry Pi Pico (A2) inputs.

Recommendation

Add 10k pull-up resistors to the 3.3v_Safty rail for nets /SaftyProcessor/thermoDrdy and /SaftyProcessor/thermoFault.

Affected Components

U13, A2

Affected Nets

/SaftyProcessor/thermoDrdy, /SaftyProcessor/thermoFault

Datasheet Evidence

MAX31856.pdf (p.7)

"3.3V --- R4 (330) --- D1 LED --- FAULT"

6. [U5] Missing Pull-up on Optocoupler Output

Full Description

The collector output of optocoupler U5 (Pin 4) is connected to GPIO35 of the ESP32-S3 without a pull-up resistor. When the phototransistor is in the OFF state, the input net will be floating, leading to unreliable data reception.

Recommendation

Add a 4.7kΩ to 10kΩ pull-up resistor from net '/MainController/DataToSafty' to the 3.3V_Main rail.

Affected Components

U5, U18

Affected Nets

/MainController/DataToSafty

Datasheet Evidence

TCMT1109.pdf (p.1)

"The TCMT110. series is a single-channel optocoupler that uses a gallium arsenide infrared-emitting diode optically coupled to a phototransistor."

7. [U4] Interface Signal Direction Mismatch

Full Description

The net /MainController/DataFromSafty is connected to the Anode (Pin 1) of optocoupler U4, identifying it as an output from the ESP32-S3. However, the name implies it is an input from the safety subsystem. This suggests a likely cross-connection (TX/RX swap) at the isolation boundary.

Recommendation

Verify the intended signal flow. If GPIO21 is meant to receive data, it must be connected to the phototransistor collector (Pin 4) rather than the LED anode.

Affected Components

U4, U18, R17

Affected Nets

/MainController/DataFromSafty

Datasheet Evidence

TCMT1109.pdf (p.1)

"Pin 1: Anode (A), Pin 2: Cathode (C), Pin 3: Emitter (E), Pin 4: Collector (C)"

8. [A2] Conflict of Parallel 3.3V Power Sources

Full Description

The '3.3v_Safty' rail is powered simultaneously by the Raspberry Pi Pico's internal SMPS (Pin 36) and an external LM1117 regulator (U17). Directly tying active outputs together can cause current fighting, instability, and thermal damage.

Recommendation

Select one source to power the rail. If redundancy is required, use Schottky OR-ing diodes to isolate the two 3.3V outputs.

Affected Components

A2, U17

Affected Nets

3.3v_Safty

Datasheet Evidence

RaspberryPi_Pico.pdf (p.4)

"Pico IO is powered from the on-board 3.3V rail and is therefore fixed at 3.3V."

9. [Q4] MOSFET Pinout Mismatch in SOT-223 Package

Full Description

The netlist connects the control signal to Pin 2 and the load to Pin 1 of the BSP296N (SOT-223). Standard SOT-223 N-channel MOSFETs follow a Gate-Drain-Source (1-2-3) pinout. Connecting the gate to Pin 2 (Drain) and the load to Pin 1 (Gate) renders the driver inoperable.

Recommendation

Verify the SOT-223 symbol footprint mapping. Correct the connections to follow the Gate=1, Drain=2/4, Source=3 convention.

Affected Components

Q4

Affected Nets

Net - (D26-A2), Net - (Q4-G)

Datasheet Evidence

BSP296NH6327XTSA1.pdf (p.8)

"SOT223 ... Pin 1, 2, 3 [Diagram shows standard 3-pin configuration]"

Warnings (10)

1. [U12] LDO Output Capacitor ESR Stability Risk

Full Description

U12 (LM1117-3.3) is connected to a 10uF ceramic capacitor (C24). Ceramic capacitors typically have ESR values below 0.1 Ω , which violates the LM1117 requirement of 0.3 Ω to 22 Ω ESR for loop stability, potentially leading to voltage oscillation.

Recommendation

Replace C24 with a tantalum capacitor (which has higher intrinsic ESR) or add a small series resistor (e.g., 0.5 Ω) to the ceramic capacitor to provide the necessary damping.

Affected Components

U12, C24

Affected Nets

3.3V_Main

Datasheet Evidence

LM1117T-3.3.pdf (p.15)

"The ESR of the output capacitor should range between 0.3 Ω to 22 Ω 10 μ F (tantalum capacitor) is required for stability."

2. [U17] LDO Stability Violation on Safety Rail

Full Description

Similar to U12, regulator U17 (LM1117-3.3) uses a 10uF ceramic capacitor (C51), risking instability due to insufficient ESR.

Recommendation

Replace C51 with a 10uF tantalum capacitor or add a series ESR resistor (0.5 Ω to 1 Ω).

Affected Components

U17, C51

Affected Nets

3.3v_Safety

Datasheet Evidence

LM1117T-3.3.pdf (p.14)

"The ESR of the output capacitor should range between 0.3 Ω to 22 Ω ."

3. [U13] Missing Current Limiting Resistor on BIAS Pin

Full Description

The BIAS pins of the MAX31856 converters (U13, U14, U15, U16) are connected to the thermocouple input (T-) through low-impedance ferrite beads. A short-to-ground at the external thermocouple probe would force high current from the internal BIAS generator, potentially damaging the IC.

Recommendation

Insert a 20kΩ resistor between the BIAS pin and the thermocouple return line (T-) as shown in the manufacturer's reference design to limit fault current.

Affected Components

U13, U14, U15, U16

Affected Nets

Net-(U13-BIAS), Net-(U14-BIAS), Net-(U15-BIAS), Net-(U16-BIAS)

Datasheet Evidence

MAX31856.pdf (p.10)

"R1 | 1 | 20kΩ ±1% resistor (0603) [connected to BIAS pin]"

4. [A2] Potential ADC Over-voltage from Current Monitoring

Full Description

The current conditioning circuit can produce peaks near 14V at the burden resistors (R2, R3, R26). With the current R4/R6 divider ratio (approx. 0.98x), the voltage at the MCU pin will exceed logic levels. This relies on internal ESD diodes to clamp the signal; given the SS16FP diodes have a Vf of 0.7V, high-current transients may cause significant injection current into the MCU.

Recommendation

Redesign the output voltage divider (e.g., using a 10k/2.2k ratio) or add a 3.3V Zener clamp to protect the Pico ADC inputs.

Affected Components

R4, R6, R5, R13, R29, R30

Affected Nets

/SaftyProcessor/Current1, /SaftyProcessor/Current2, /SaftyProcessor/Current3

Datasheet Evidence

SS16FP.pdf (p.1)

"V_F (max) at 1.0A = 0.70V"

5. [U18] Pin Conflict on Octal SPI Modules

Full Description

GPIO35, GPIO36, and GPIO37 are used for external signals. On ESP32-S3-DevKitC-1 boards with Octal SPI flash/PSRAM, these pins are used for internal communication and are unavailable for external use.

Recommendation

Verify that the intended hardware is a standard Quad SPI (WROOM-1) module. If an Octal SPI module is used, relocate these signals to GPIO33-34 or GPIO38-42.

Affected Components

U18

Affected Nets

/MainControler/DataToSafty, /MainControler/Fault, /MainControler/thermoFault_0

Datasheet Evidence

ESP32-S3-DevKitC-1 v1.1 - ESP32-S3 - — esp-dev-kits latest documentation.pdf (p.3)

"For boards with Octal SPI flash/PSRAM memory embedded... the pins GPIO35, GPIO36 and GPIO37 are used for the internal communication... thus not available for external use."

6. [A2] Non-Standard Power Input on Pico 3V3 Pin

Full Description

The 3.3v_Safty rail is connected to Pin 36 (3V3) while VSYS (Pin 39) is floating. The datasheet specifies Pin 36 as an output of the onboard regulator, not a power input, which may lead to back-powering issues.

Recommendation

Power the Pico via the VSYS pin (Pin 39) with a 1.8V to 5.5V source to allow the onboard regulator to function normally.

Affected Components

A2

Affected Nets

3.3v_Safty

Datasheet Evidence

RaspberryPi_Pico.pdf (p.12)

"3V3 pin (pin 36) is the output of the on-board SMPS and is the main 3.3V supply to the RP2040 and its I/O; it is not listed as a power input pin."

7. [A2] Floating ADC_VREF on Safety Processor

Full Description

ADC_VREF (Pin 35) is left floating, which may introduce noise or offset errors into current monitoring measurements (Current1/2/3).

Recommendation

Connect ADC_VREF to the 3.3V rail or an external reference to ensure ADC accuracy for safety-critical current monitoring.

Affected Components

A2

Datasheet Evidence

RaspberryPi_Pico.pdf (p.20)

"3.3V and ADC_VREF are shorted together (which is a valid thing to do if you don't care about noise and want to reduce the inherent offset)."

8. [J12] Missing ESD Protection on User-Facing Terminals

Full Description

Thermocouple connectors J11, J12, J13, and J15 lack dedicated ESD protection. These terminals are vulnerable to discharge from user handling or induced transients on long thermocouple leads.

Recommendation

Add bidirectional TVS diodes (e.g., 5V rating) to Ground on each signal line near the terminal entry point.

Affected Components

J12, J13, J11, J15

Affected Nets

Net - (FB3-Pad1), Net - (FB4-Pad1), Net - (FB5-Pad1), Net - (FB6-Pad1)

Datasheet Evidence

MAX31856.pdf (p.1)

"The MAX31856 evaluation system provides the hardware and software necessary to evaluate the MAX31856."

9. [R57] Redundant Parallel Pull-up Resistors on SPI Buses

Full Description

Both the Main and Safety SPI buses contain redundant pull-up resistors (e.g., R57/R62, R31/R51) on shared lines (CLK, MOSI, CS), resulting in an effective 5kΩ pull-up. This increases BOM count without functional benefit.

Recommendation

Remove one resistor from each parallel pair to simplify the BOM.

Affected Components

R57, R62, R58, R63, R31, R51, R32, R52, R33, R53

Affected Nets

/MainController/CLK, /MainController/MOSI, /SaftyProcessor/CLK, /SaftyProcessor/CS0, /SaftyProcessor/MOSI

Datasheet Evidence

MAX31856.pdf (p.7)

"MAX31856PMB1 Schematic [shows only one set of pull-up resistors if used]"

10. [U18] Unconnected Chip Select Line (CS3)

Full Description

Net /MainController/CS3 is connected to pull-up R20 but is not routed to any GPIO pin on the ESP32-S3 (U18).

Recommendation

Connect /MainController/CS3 to an available GPIO on U18 if it is intended to be software-controlled.

Affected Components

U18, R20

Affected Nets

/MainController/CS3

Datasheet Evidence

ESP32-S3-DevKitC-1 v1.1 - ESP32-S3 - — esp-dev-kits latest documentation.pdf (p.3)

"GPIO pins are broken out to the pin headers on the board."

Missing Information (4)

[IC1] Buck Converter LT8631 Specifications Unverified

Verification of feedback reference voltage, inductor connectivity requirements, and switching frequency was not possible without the LT8631 datasheet.

Information Needed

Manufacturer datasheet for LT8631 to verify the feedback divider (R47/R48), RT frequency (R46), and IND pin internal logic.

[IC2] Safety Buck Regulator Parameters Unverified

The exact undervoltage lockout thresholds (EN/UV divider) and feedback voltage for the safety buck converter could not be confirmed.

Information Needed

Manufacturer datasheet for LT8631 to verify EN/UV pin thresholds and the safety 5V rail feedback network.

[C7, C9, C11] Capacitor DC Bias Derating Unverified

The current monitor filter capacitors are rated for 16V with expected signal peaks near 14V. High capacitance loss is expected but cannot be quantified without exact capacitor part numbers.

Information Needed

Detailed manufacturer data for C7, C9, and C11 to verify capacitance at 14V DC bias.

[R85, R43] Non-Functional Zero-Ohm Resistor Layout

Zero-ohm resistors R85 and R43 have both pins connected to the same net in the netlist, effectively shorting them out.

Information Needed

Clarification of whether these were intended as configuration jumpers or net-ties between distinct supply planes.

Verified Items (155)

Components (111)

J10	D22	C12	C13
C17	C18	C22	C23
D35	J19	C14	C48
C49	R79	R80	R87
R88	R19	R21	R22
R23	R24	R25	R27
R28	R34	R77	R78
U6	D25	D26	J5
R12	R14	R15	R16
R17	R18	C10	U14
U15	C33	C34	C38
C39	R56	R61	C30
C31	C32	C35	C36
C37	FB3	FB4	FB5
FB6	U16	C28	C29
C43	C44	R51	R70
J11	J15	Q1	Q2
D19	D21	R36	R37
R40	R41	J8	J9
J2	J3	J4	J6
J7	J18	R2	R3
R4	R5	R6	R13
R26	R29	R30	D3
D4	D5	D6	D7
D8	D27	D28	D29
D30	D31	D32	C7
C9	C11	D18	D20
D23	D33	C15	

Nets (44)

12V_Main	3.3V_Main	GND_Main	12v_Safty
GND_Safty	3.3v_Safty	5V_Main	/MainControler/CLK
/MainControler/MOSI	/MainControler/MISO	/MainControler/CS0	/MainControler/CS1
/MainControler/CS2	/SaftyProcessor/estop	/SaftyProcessor/PicoTx	/SaftyProcessor/PicoRx
/SaftyProcessor/fault	/MainControler/thermoDrdy_0	/MainControler/thermoFault_0	/MainControler/thermoDrdy_1
/MainControler/thermoFault_1	/SaftyProcessor/CS0	Net-(U13-T+)	Net-(U13-T-)
Net-(U16-T+)	Net-(U16-T-)	Net-(Q1-G)	Net-(Q2-G)
Net-(J4-TIP)	Net-(J4-SLEEVE)	Net-(J6-TIP)	Net-(J6-SLEEVE)
Net-(J18-TIP)	Net-(J18-SLEEVE)	/SaftyProcessor/Current1	/SaftyProcessor/Current2
/SaftyProcessor/Current3	/MainControler/thermoDrdy_2	/MainControler/thermoFault_2	/SaftyProcessor/CLK
/SaftyProcessor/MISO	/SaftyProcessor/MOSI	/SaftyProcessor/thermoDrdy	/SaftyProcessor/thermoFault